

$I(\mathcal{C}_t; o_{t+1:\infty} \mid a_{t:\infty})$
full-history predictive information

retained by M_t

residual lost
given M_t

$I(\mathcal{C}_t; o_{t+1:\infty} \mid M_t, a_{t:\infty})$
information full history still has beyond model

$$S(M_t) = 1 - \frac{\text{residual predictive information after } M_t}{\text{total predictive information in } \mathcal{C}_t}$$

Undefined when the denominator is zero: no prediction task, no fraction retained.